

Standalone A-to-B Navigation with G1 EDU in Gazebo

Standalone Point-to-Point Navigation Benchmark for Indoor Robot Operation

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Research Goals

- Develop a reliable indoor A-to-B navigation Cognitive Architecture for the Unitree G1 EDU in Gazebo.
- Configure a complete ROS2 Nav2 pipeline
- Compare DWB, MPPI, and RPP local planners
- Test five indoor benchmark scenarios
- Evaluate success, time, path efficiency, collisions, recovery, and goal accuracy
- Identify the most suitable planner for the robot platform

Main Challenge

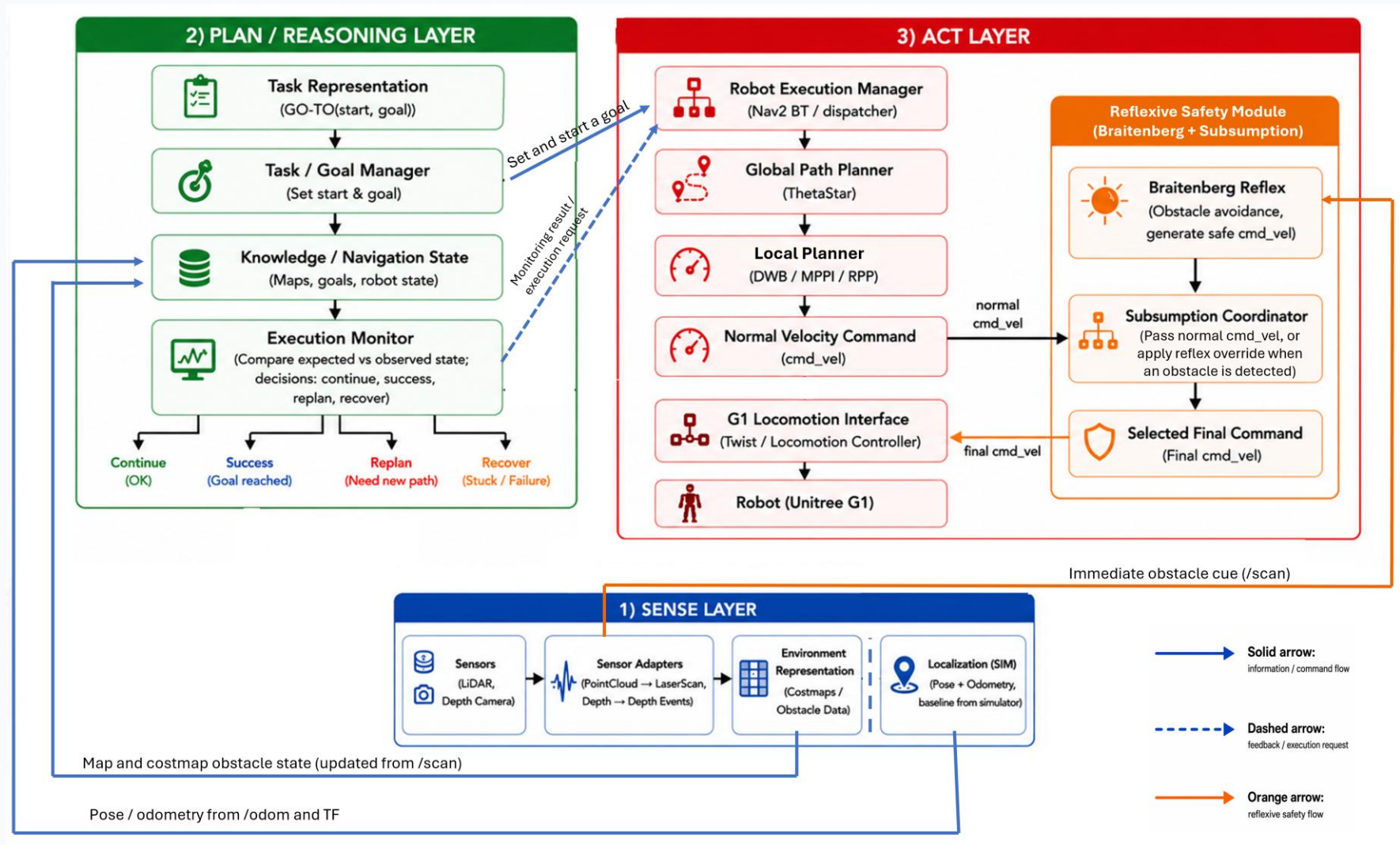
The Unitree G1 is not a small differential-drive robot in spirit: it has a larger footprint, a humanoid body with 29 DOF structure, and must navigate through constrained indoor layouts without collisions.

Evaluation Setup

A complete simulation pipeline: known map, simulator localization, global path planning, local obstacle avoidance, behavior recovery, and run-to-run repeatability.



Hybrid Reactive–Deliberative Cognitive Architecture



Hybrid Reactive–Deliberative Cognitive Architecture

Hierarchical navigation with execution monitoring and fast bio-inspired safety

Hierarchical Navigation — Deliberative Path

- The navigation objective is represented as a **GO-TO(start, goal)** task .
- The Robot Execution Manager coordinates **ThetaStar global planning** and the selected local planner: **DWB, MPPI, or RPP**.
- Execution is continuously monitored using the observed pose, progress, clearance, controller status, and goal state.
- Based on the monitoring result, the system may **continue, replan, recover, succeed, or fail**.

Bio-inspired Reflexive Safety — Reactive Path

- Immediate obstacle information activates a **Braitenberg-inspired reflex**.
- The reflex generates a short-term safe velocity command without waiting for full replanning.
- Through priority arbitration, the safety command can temporarily override the normal velocity command.

The architecture combines a hierarchical deliberative navigation path with a fast reflexive safety pathway.

ROS 2 implementation mapping

How the cognitive architecture maps to concrete topics, actions, services, and TF

Sense and observed state

- The 3D LiDAR data are adapted into a two-dimensional `/scan` stream for navigation and obstacle detection.
- The Global Costmap combines the static map with obstacle updates, while the Local Costmap represents nearby obstacles for local motion control.
- `/odom` and the TF chain provide the robot pose and observed motion state.
- The Benchmark Logger records `/odom`, `/cmd_vel`, `/scan`, and `/reflex_active` to compute trajectory, clearance, motion, and reflex metrics.

Planning, control and safety

- Navigation goals are sent through the long-running Nav2 `/navigate_to_pose` action.
- The Global Planner uses ThetaStar to generate the route, while the selected Local Planner—DWB, MPPI, or RPP—produces the motion command.
- The Local Planner publishes the normal command on `/nav_cmd_vel`. The subsumption coordinator forwards it during normal operation or selects `/reflex_cmd_vel` when the reflex is active. The final command is published on `/cmd_vel`.
- Services handle discrete operations such as enabling the logger and spawning or deleting the dynamic obstacle.

ROS communication roles



Topics

Continuous sensor, state, and command streams



Action

`/navigate_to_pose`



Services

discrete operations



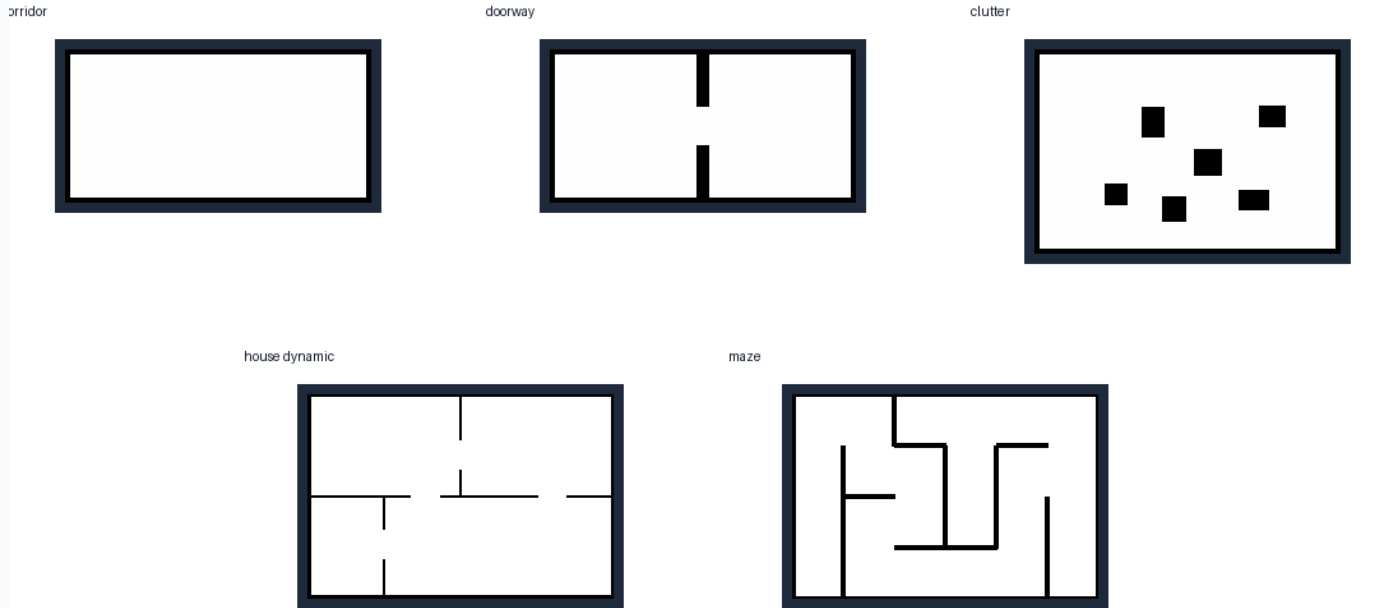
TF

`map` → `odom` → `base_link`

Key idea: ROS 2 keeps the cognitive layers implementable through topic streams, action feedback, service calls, and TF frames.

Benchmark scenarios and metrics

Five indoor tasks stress different navigation behaviors.



Scenario roles

corridor = baseline timing; doorway = narrow passage; clutter = local obstacle adaptation; house_dynamic = moving blockage; maze = high-curvature route.

Metrics

success rate, completion time, collision count, recovery count, reflex count, final goal error.

Run matrix

5 scenarios × 3 planners × 3 repetitions = 45 runs. Each run writes telemetry, per-run summary CSV, aggregate CSV and plots.

Benchmark scenarios and metrics

Corridor Scenario

DWB

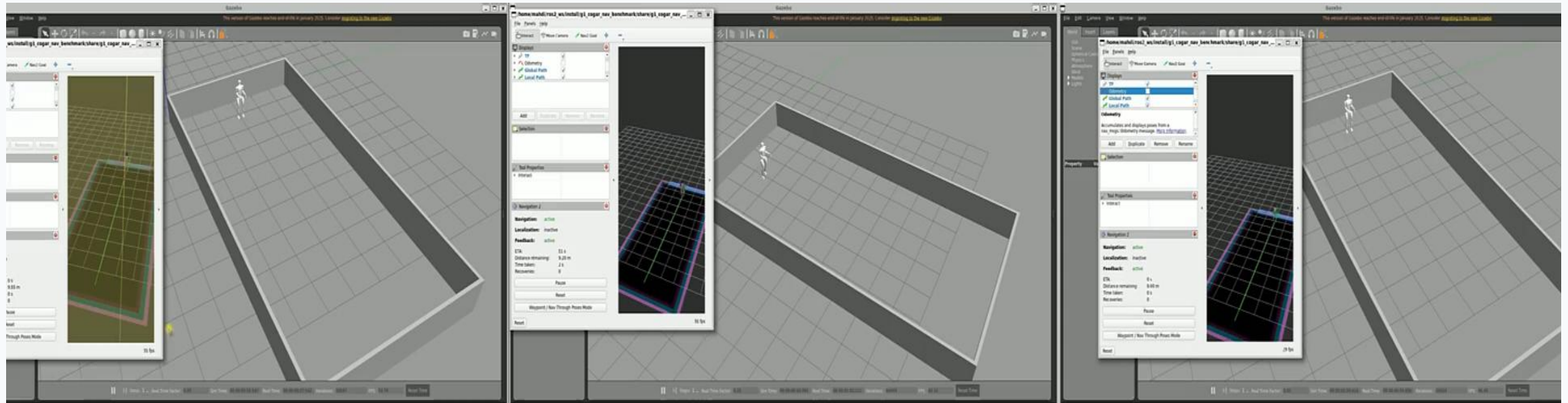
planner

MPPI

planner

RPP

planner



Benchmark scenarios and metrics

Doorway Scenario

DWB

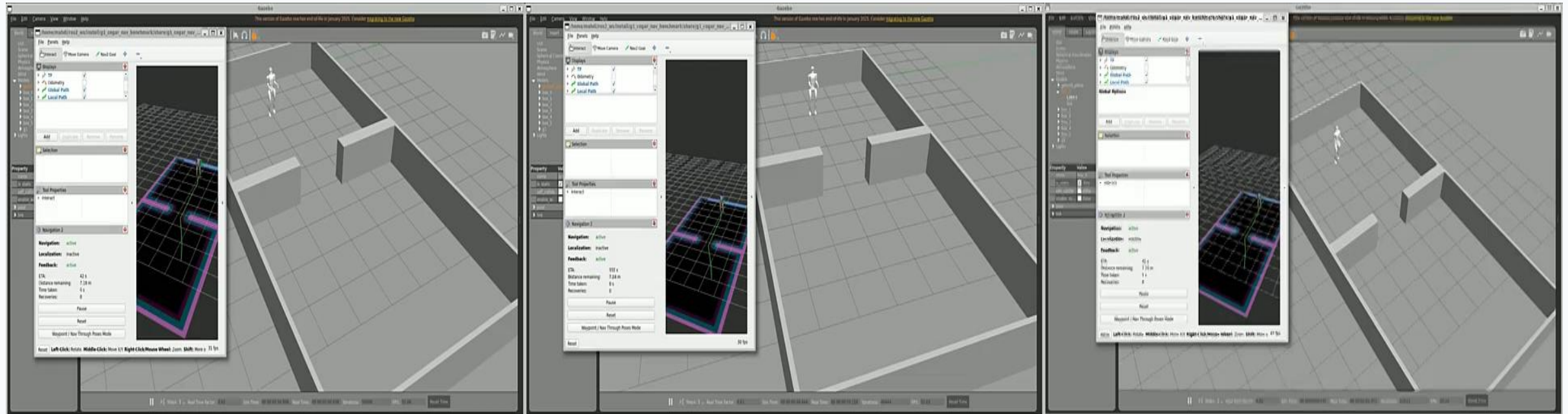
planner

MPPI

planner

RPP

planner



Benchmark scenarios and metrics

Clutter Scenario

DWB

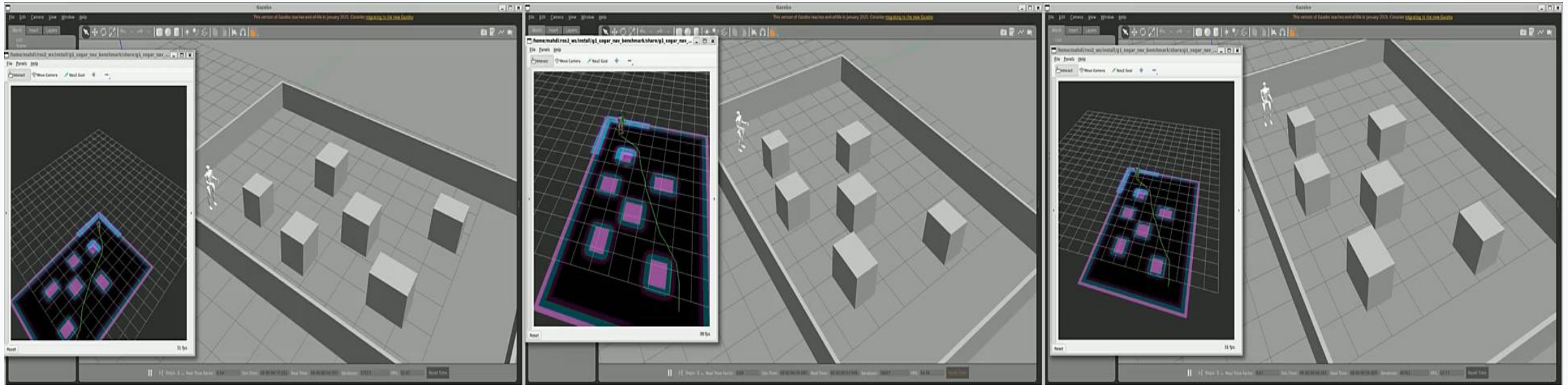
planner

MPPI

planner

RPP

planner



Benchmark scenarios and metrics

Dynamic Blockage Scenario

DWB

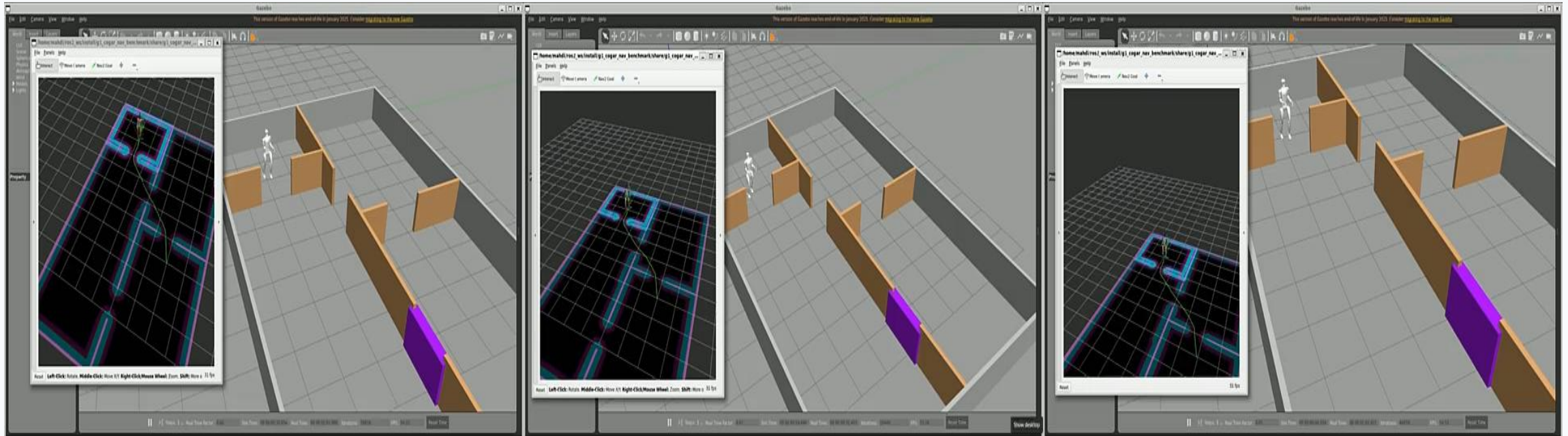
planner

MPPI

planner

RPP

planner



Benchmark scenarios and metrics

Maze scenario

DWB

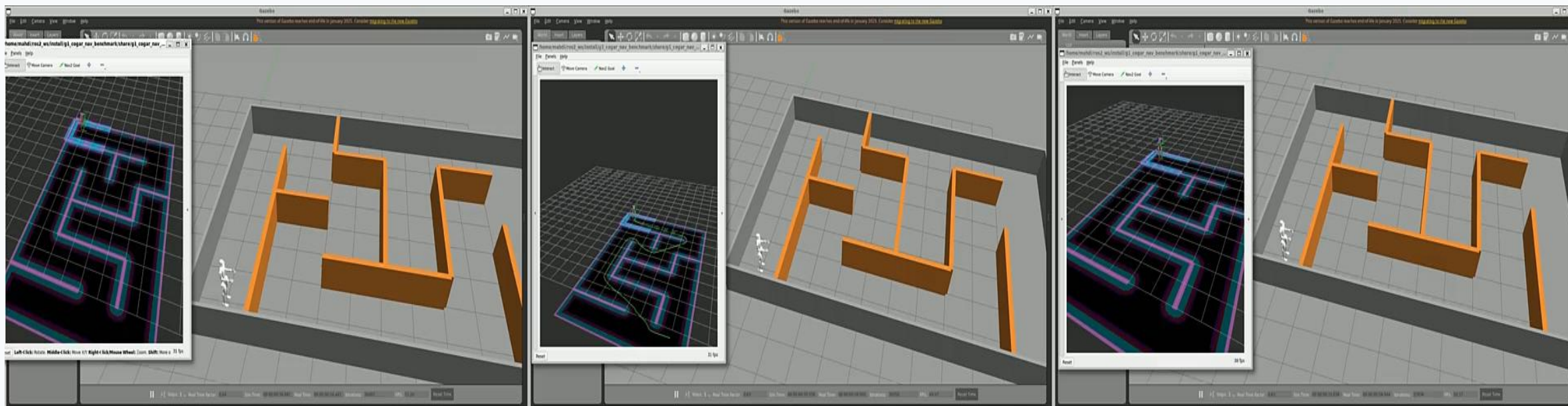
planner

MPPI

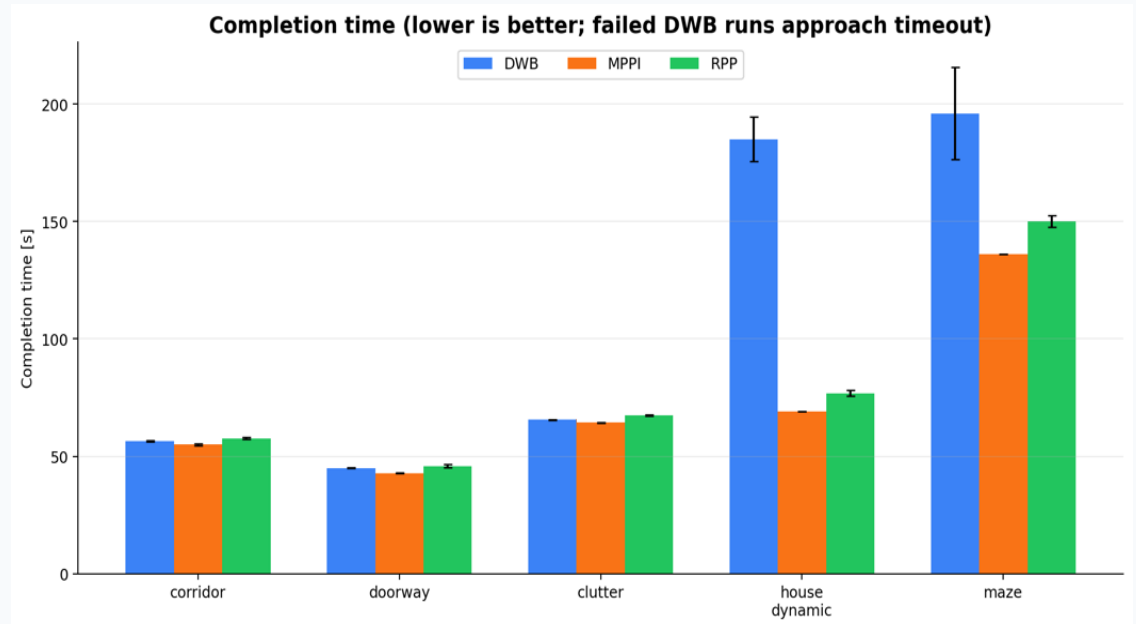
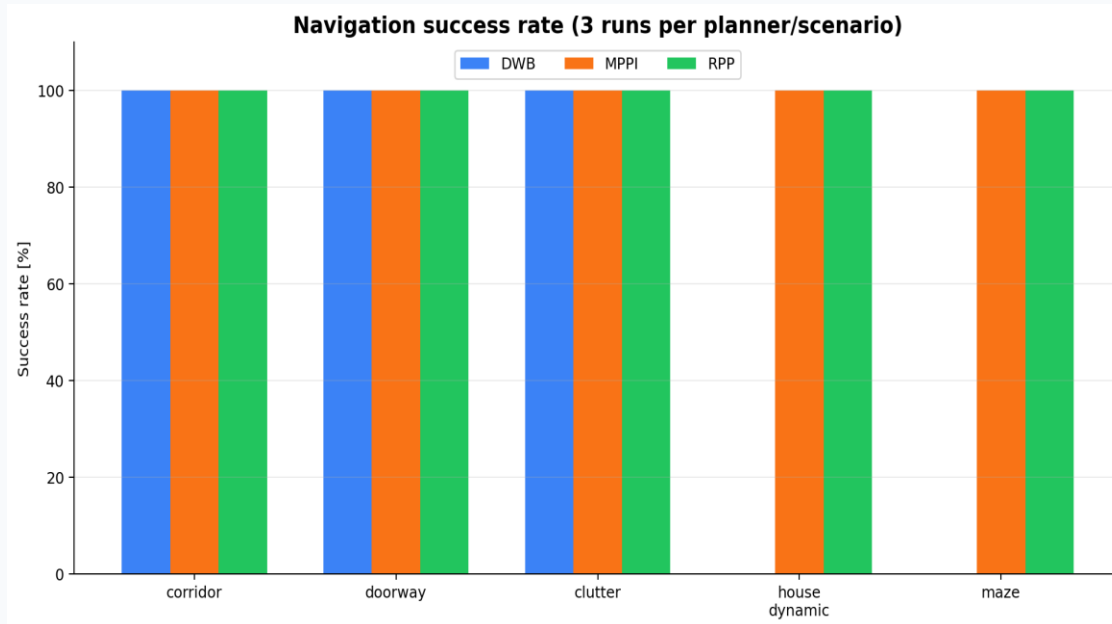
planner

RPP

planner



Results I: success and completion time



Main observation

DWB reached all simple tasks but failed in house_dynamic and maze. MPPI and RPP completed all tasks.

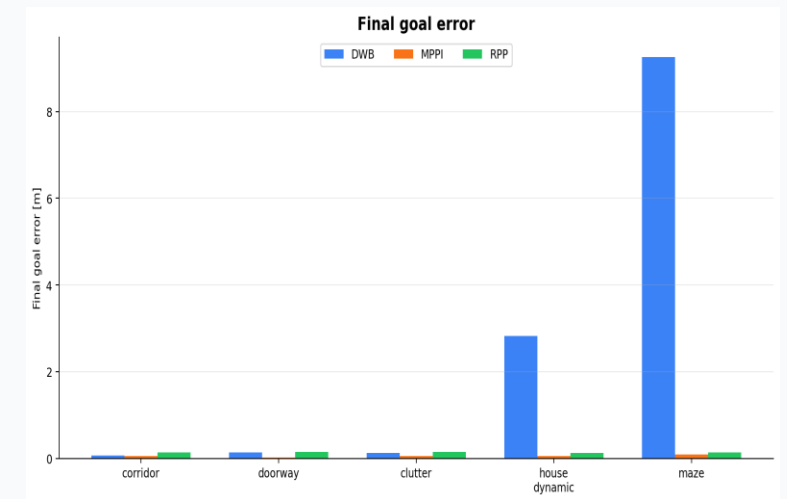
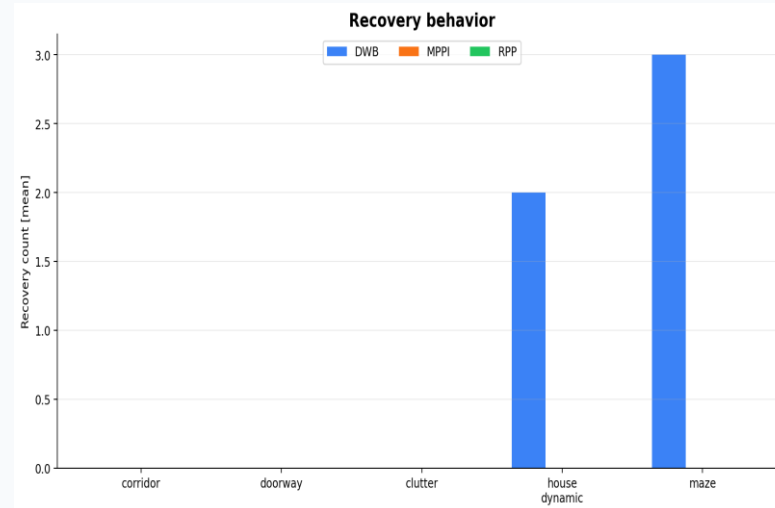
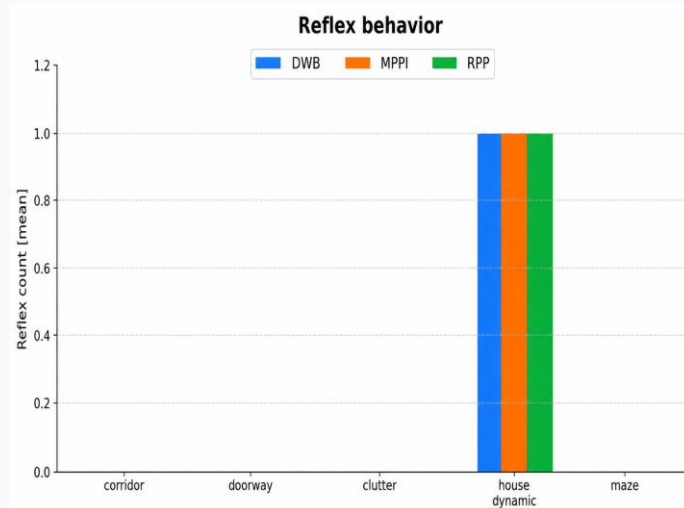
Timing

MPPI was generally the fastest among the successful planners in the complex scenarios.

Interpretation

The results match the planner properties: predictive MPPI handles complex local decisions better than other local planners.

Results II: recovery, goal error and reflex



Goal error

MPPI showed the smallest final goal error in most successful tasks. RPP often stopped safely but less precisely.

Recovery

DWB triggered recovery in the difficult scenarios. MPPI/RPP usually completed the difficult scenarios through normal navigation and replanning, without invoking Nav2 recovery.

Reflex

Braitenberg briefly overrode `cmd_vel` in `house_dynamic` to avoid the sudden obstacle.

Safety

No collision-proxy events were recorded

Note on recovery_success

`Recovery_success` defaults to 1, even when no recovery is executed. Actual recovery activation is indicated by `recovery_count > 0`; a value of 0 in `Recovery_success` means recovery failed to restore progress, as for DWB in `house_dynamic` and `maze`.

Planner-by-planner interpretation

Planner	Success	Mean time	Mean recovery	Mean final error
DWB	60% avg	109.3s	1.0	2.48m
MPPI	100% avg	73.5s	0.0	0.05m
RPP	100% avg	79.6s	0.0	0.14m

DWB: baseline

Good in corridor, doorway and clutter.
Weak in maze and dynamic-obstacle conditions under this configuration. The failures are visible in success rate, recovery and final error.

MPPI: best overall

The most consistent profile in this benchmark: 100% success, strong completion time, low final error, and no collisions.

RPP: robust but conservative

Reached the goal in all scenarios, but generally required more time and had slightly larger final error than MPPI.

Congruence of results and conclusions

Supported conclusion 1

In the tested Gazebo scenarios, MPPI is the best overall planner profile for this G1 EDU navigation pipeline.

Supported conclusion 2

RPP is robust and conservative, but its regulated behavior makes it slower and slightly less precise at the final pose.

Supported conclusion 3

DWB is suitable as a simple baseline, but not robust enough for the hardest maze and dynamic cases with my tuning.

Limitations

Simulation-only evaluation, only three repetitions, and simplified humanoid locomotion without full-body dynamics.

TEB note

TEB was suggested in the assignment as an example, but I used RPP as the third planner because a stable ROS2 Humble TEB profile was not available in my environment.

What I would improve

Increase the number of experimental repetitions, test the system on the real G1 and in dynamic-human scenarios, and perform parameter sensitivity analysis. With stronger computing hardware, the benchmark could also be extended to full-body dynamic simulation with balance control, gait generation, and more realistic humanoid locomotion for the 29-DOF robot.

Final takeaway

The implemented system integrates hierarchical navigation, execution monitoring, reflexive safety, and quantitative evaluation.

ROS2

modular package

Nav2

planner profiles

Gazebo

benchmark worlds

Conclusion:

MPPI is the recommended local planner for the tested G1 Gazebo benchmark. RPP is a robust but more conservative alternative, while DWB remains a useful baseline that reveals limitations in difficult, high-curvature scenarios and recovery behavior.